Name: G. Pavan
Reg No: 192565011
ASSESSMENT TOOL-3 SLOT: C
Course
CLASS TEST Code: CSA1603

1] Data mining & KDD

Datamining:- Process of extracting useful Patterns and knowledge from large datasets.

Two Tasks:

- Classification
- Clustering

KDD Steps:

1. Data Selection
2. Data cleaning
3. Data Integration
4. Data transformation
5. Datamining
6. Pattern Evaluation
7. Knowledge Presentation

Role of Preprocessing:- Improves data quality, removes noise, and increases mining accuracy

2] Data cleaning

Definition:- Process of removing errors, missing values, and inconsistencies from data.

Two Data quality Issues:-

- missing values
- Noisy data

Handling missing values:-

- Replace with mean/median/mode
- Ignore me record

Handling Noisy Data:

- Binning
- Regression

3) Data integration

Definition:- Combining data from multiple sources into one dataset.

challenges:

- Redundancy
- Inconsistency

Handling:

- Remove duplicate data
- Standardize formats and naming

4. Covariance analysis

Definition: Measures how two variables change together

Formulas:-

Population:

$$Cov(x, y) = \frac{\Sigma(x - \mu_x)(y - \mu_y)}{N}$$

Sample:

$$Cov(x, y) = \frac{\Sigma(\bar{x} - x)(\bar{y} - y)}{n - 1}$$

5. Stocks A & B

Covariance = Positive

Answer:- The stock Prices rise together and are not independent.

6. Datamining Architecture

Components:

- Data Sources
- Data cleaning & Integration
- Data warehouse
- Database Server
- Data mining Engine
- Pattern Evaluation
- Knowledge Base
- User Interface

These Components work together to discover useful knowledge from data

7] Covariance

Given

$X = \{4, 5, 6, 7, 9\}$

$Y = \{8, 3, 7, 5, 6\}$

Covariance $= -0.45$

Answer: Negative Covariance means the variables move in opposite directions.

8. Data Pre Processing steps

1. Data cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization

These steps improve data quality before mining.

9. Equal-frequency Binning

Definition :- Divides data into bins with an equal number of values.

for 27 values and 3 bins

- Bin 1: first 9 values
- Bin 2: Next 9 values
- Bin 3: last 9 values

Replace each bin by it means (local smoothing).

10. Data Discretization

Definition: Converts continuous data into intervals (bins).

Importance:

- Reduces noise
- Simplifies analysis
- Improves mining accuracy

Equal-width: Equal interval size.

Equal-frequency: Equal number of values in each bin

Comparison:- Equal-width is simple; equal-frequency handles skewed data better.